Experiment No. 6
Identify and deal with Outliers on the given dataset and evaluate the performance
Date of Performance: 10/03/2025
Date of Submission: 17/03/2025

Aim: Identify and deal with Outliers on the given dataset and evaluate the performance.

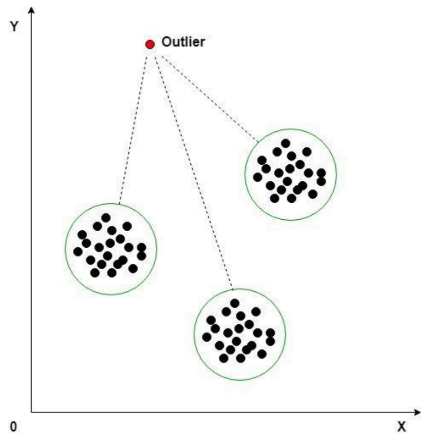
Objective: Able to identify the outliers and deal with them.

Theory:

An outlier is an object that deviates significantly from the rest of the objects. They can be caused by measurement or execution error. The analysis of outlier data is referred to as outlier analysis or outlier mining.

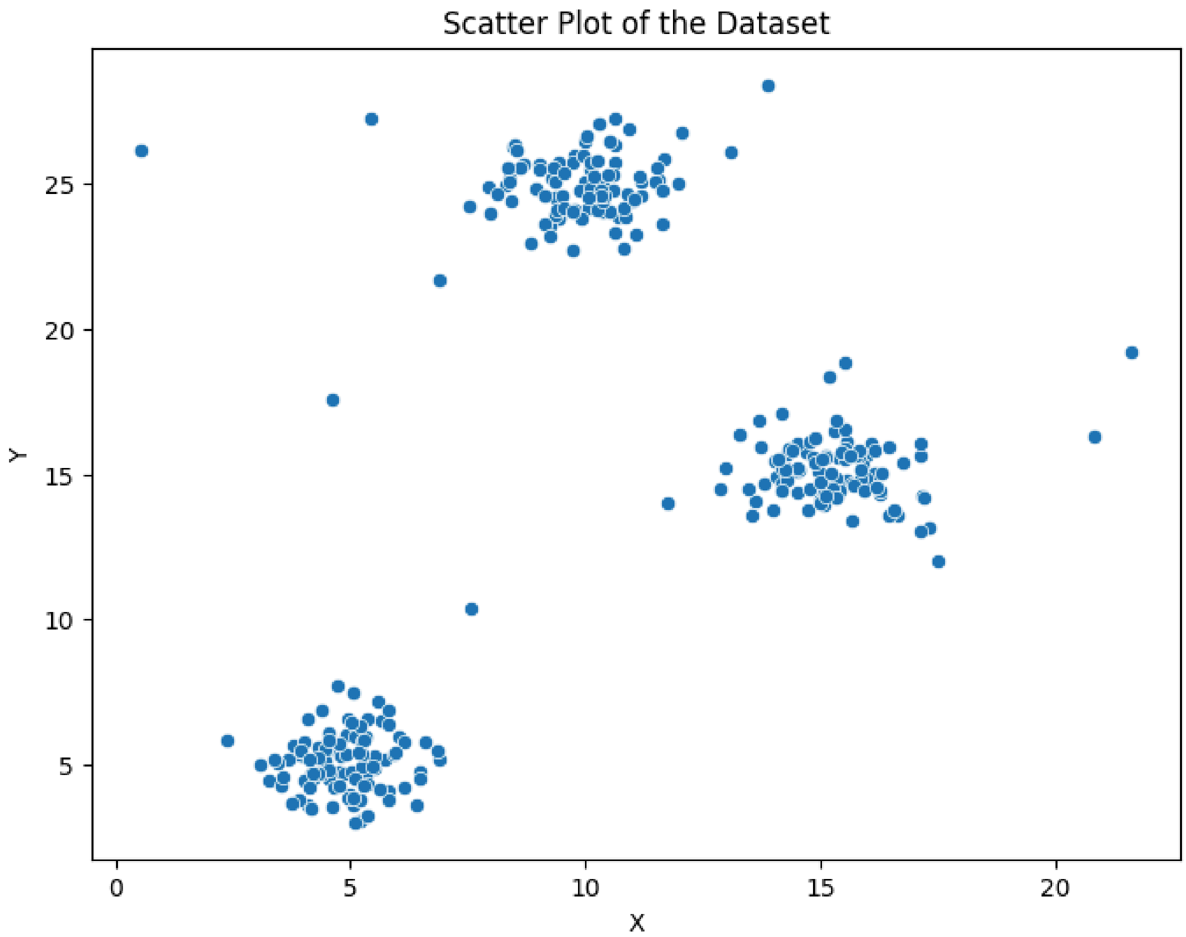
Why outlier analysis?

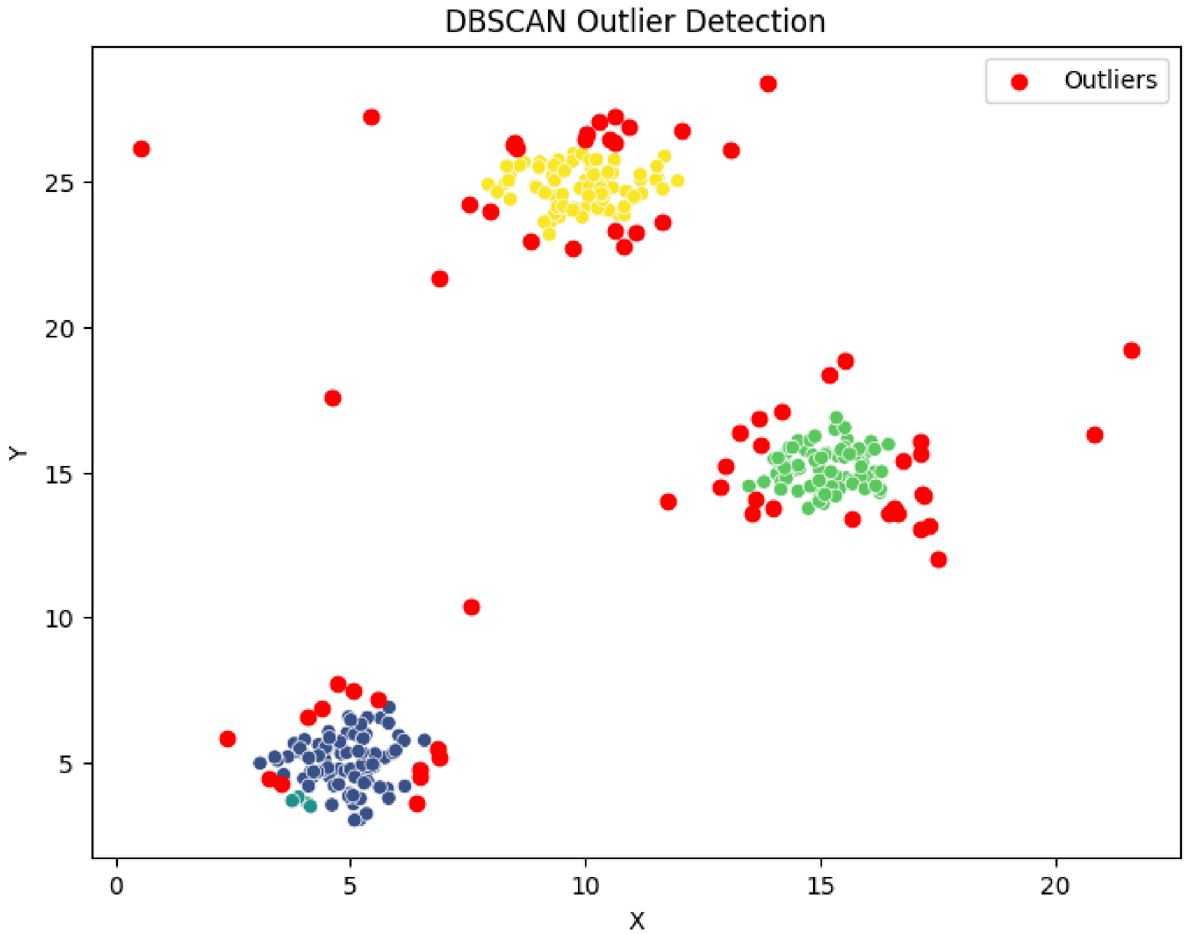
Most data mining methods discard outliers noise or exceptions, however, in some applications such as fraud detection, the rare events can be more interesting than the more regularly occurring one and hence, the outlier analysis becomes important in such case.



Code:
https://colab.research.google.com/drive/1H93v0C90tLKb8jsHXfRFt2H6TeQlYTBS?usp=sharing

Output:





	Metric	Before Outlier Removal	After Outlier Removal
0	Accuracy	0.967742	1.0
1	Precision	0.923077	1.0
2	Recall	0.923077	1.0
3	F1 Score	0.923077	1.0

Conclusion:

Removing outliers significantly improved the model's performance. Before outlier removal, the model had an accuracy of **96.77%**, with precision, recall, and F1-score at **92.31%**. After removing outliers, all metrics improved to **100%**, indicating perfect classification. This improvement suggests that outliers were adding noise to the dataset, making it harder for the model to distinguish between normal and anomalous points. By eliminating these anomalies, the model could better learn the underlying patterns, leading to better generalization and reliability. However, in real-world applications, removing outliers should be done carefully, as some anomalies may contain valuable insights, such as fraud detection or rare events.